Xinbao Qiao

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Education

The Chinese University of Hong Kong Ph.D., Information Engineering. Advisor: Angela Yingjun Zhang. *Hong Kong. 2026~present*

Research Areas My research primarily studies lifecycle management of data in AI models, focusing on theoretical methods and practical problems that arise as data are generated, used, and deleted.
Teaching AI Algorithms and Systems. Secure Artificial Intelligence.

Zhejiang University M.Sc. in Artificial Intelligence. Advised by Meng Zhang. *Zhejiang. Sep 2022 ~ Dec 2025*
Shandong University B.Eng. in Communication Engineering. *Shandong. Sep 2018 ~ Jul 2022*

Experience

Doctoral Research @ The Chinese University of Hong Kong *Hong Kong. 2026 ~ present*
- Study data-centric ML and AI-network co-design, including distributed Wasserstein barycenter methods for shared references without pooling raw data. Paper #5

Research on Data-Centric ML Systems @ Zhejiang University *Zhejiang. Mar 2023 ~ Dec 2025*
- Developed soft-weighted unlearning methods that continuously correct data influence for fairness and robustness without binary erasure. Paper #2
- Designed Hessian-free online certified unlearning methods using recollected trajectory statistics instead of explicit Hessian inversion, supporting online deletion requests without second-order solves. Paper #3
- Studied exact and efficient random-forest unlearning in dynamic online environments, preserving predictive utility under incremental data updates. Paper #4

Research on Trustworthy LLM Systems @ NUSRI CQ *Chongqing. Jun 2025 ~ Dec 2025*
- Analyzed low-resource sample selection in recursive synthetic-data training and proposed collaborative Wasserstein evaluation proxies. Paper #1
- Examined illusory pattern perception in LLMs and developed an interpretable framework for spurious inference behavior and misleading responses. Paper #6

Open-Source Contributions and Services

- **Research code releases**, public code for certified unlearning and model-collapse work.
- **Xinbaopedia**, public academic homepage and wiki-style research archive with papers, figures, CV, and project notes.
- **Peer-Reviewing**, reviewer for ICML 2026, NeurIPS 2025 and 2026, ICLR 2025, AAAI 2025 and 2026, and IEEE TNNLS.

Selected Publications

Asterisks (*) denote co-first authorship. Accepted papers are listed before under-review manuscripts.

1. **When Sample Selection Bias Precipitates Model Collapse** *ICML 2026.*
   Xinbao Qiao, Xianglong Du, Wei Liu, Jingqi Zhang, Peihua Mai, Meng Zhang, Yan Pang
   Low-resource verification can turn local sample selection into persistent tail pruning.
2. **Beyond Binary Erasure: Soft-Weighted Unlearning for Fairness and Robustness** *AAAI 2026.*
   Xinbao Qiao, Ningning Ding, Yushi Cheng, Meng Zhang
   Soft-weighted corrective unlearning for non-binary fairness and robustness interventions.
3. **Hessian-Free Online Certified Unlearning** *ICLR 2025.*
   Xinbao Qiao, Meng Zhang, Ming Tang, Ermin Wei
   Certified unlearning without explicit Hessian inversion through recollected statistics.
4. **DynFrs: An Efficient Framework for Machine Unlearning in Random Forest** *ICLR 2025.*
   Shurong Wang, Zhuoyang Shen, **Xinbao Qiao**, Tongning Zhang, Meng Zhang
   Exact and efficient random-forest unlearning in dynamic environments.
5. **Federated Learning as Optimal Transport: Barycentric Multi-Prototype Classification** *Under review.*
   Xinbao Qiao, Wenjing Yan, Ying-Jun Angela Zhang
   Barycentric multi-prototype classification for federated representation geometry.
6. **Illusory Pattern Perception Drives Spurious Inference in Large Language Models** *Under review.*
   Peihua Mai, Zhuoyan Shao, **Xinbao Qiao**, Meng Zhang, Xinyue Zhou, Yan Pang
   Analysis of spurious pattern perception and interpretable behavior in LLMs.